

Date: 10/09/2025

Lab Practical #14:

Implementation of parity bit check Using C/Java language with example.

Practical Assignment #14:

1. C/Java Program: Implementation of parity bit check Using C/Java language.

```
import java.util.Scanner;

public class ParityBitCheck {

    public static String addParityBit(String dataBits, String parityType) {
        int countOnes = 0;
        for (char bit : dataBits.toCharArray()) {
            if (bit == '1') countOnes++;
        }

        char parityBit;

        if (parityType.equalsIgnoreCase("even")) {
            parityBit = (countOnes % 2 == 0) ? '0' : '1';
        } else {
            parityBit = (countOnes % 2 == 0) ? '1' : '0';
        }

        return dataBits + parityBit;
    }

    public static boolean checkParity(String data, String parityType) {
        char parityBit = data.charAt(data.length() - 1);
        String dataBits = data.substring(0, data.length() - 1);
```

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```
int countOnes = 0;
for (char bit : dataBits.toCharArray()) {
    if (bit == '1') countOnes++;
}

boolean isValid;
if (parityType.equalsIgnoreCase("even")) {
    isValid = ((countOnes % 2 == 0) && parityBit == '0') ||
              ((countOnes % 2 == 1) && parityBit == '1');
} else {
    isValid = ((countOnes % 2 == 0) && parityBit == '1') ||
              ((countOnes % 2 == 1) && parityBit == '0');
}

return isValid;
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter binary data (without parity bit): ");
    String dataBits = scanner.nextLine();

    System.out.print("Enter parity type (even/odd): ");
    String parityType = scanner.nextLine();

    String dataWithParity = addParityBit(dataBits, parityType);
```



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```
System.out.println("Data with " + parityType + " parity bit: " +  
dataWithParity);  
  
if (checkParity(dataWithParity, parityType)) {  
    System.out.println("Parity check PASSED.");  
} else {  
    System.out.println("Parity check FAILED.");  
}  
  
scanner.close();  
}  
}
```